

Replay-Aware CVRP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: cvrp_failure_replay.
- Best CVRPLIB holdout gap: cvrp_failure_replay.
- Best synthetic holdout gap: cvrp_random_replay.

Run Metadata

- run_name: run_20260513_191859_f
- started_at_local: 2026-05-13 19:18:59
- finished_at_local: 2026-05-13 19:21:46
- duration_hhmm: 00:03
- duration_seconds: 166.97
- seed_offset: 5000
- replicate_label: f
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final CVRPLIB Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
cvrp_no_replay	none	score_only	False	0.245169	0.012431	0.14173	0.0	0.58	0.0
cvrp_random_replay	random	score_only	False	0.24255	0.005376	0.137139	0.821106	0.58	0.044396
cvrp_failure_replay	failure	score_only	False	0.241049	0.005376	0.136305	0.0	0.58	0.0
cvrp_failure_replay_compression	failure	novelty_gate	True	0.245482	0.005376	0.138768	0.0	0.58	0.0

Condition Notes

cvrp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.245169, synthetic 0.012431, combined 0.14173.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.245169 across 5 instances; family means: A=0.128272, B=0.249482, E=0.291747, P=0.306859.
- Panel synthetic_holdout mean gap 0.012431 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.028221.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 577, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.259825}, {"best_known_cost": 937, "cost": 1064, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.135539}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.24255, synthetic 0.005376, combined 0.137139.
- Accepted-epoch count 2, mean accepted code novelty 0.821106, and final complexity 0.58.
- Adaptation efficiency 0.044396 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.24255 across 5 instances; family means: A=0.128272, B=0.249482, E=0.284069, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 577, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.259825}, {"best_known_cost": 937, "cost": 1057, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.128068}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: CVRPLIB 0.241049, synthetic 0.005376, combined 0.136305.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.241049 across 5 instances; family means: A=0.116928, B=0.249482, E=0.287908, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1057, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.128068}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

cvrp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: CVRPLIB 0.245482, synthetic 0.005376, combined 0.138768.

- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.58.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 5, 'failure_cases': 5, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_cvrplib mean gap 0.245482 across 5 instances; family means: A=0.116928, B=0.259605, E=0.289827, P=0.301444.
- Panel synthetic_holdout mean gap 0.005376 across 4 instances; family means: alternating_belt=0.021505, clustered_demand=0.0, corridor_split=0.0, radial_heavy=0.0.
- Worst recent training cases: [{"best_known_cost": 458, "cost": 604, "family": "P", "name": "P-n40-k5", "optimality_gap": 0.318777}, {"best_known_cost": 937, "cost": 1057, "family": "A", "name": "A-n44-k6", "optimality_gap": 0.128068}, {"best_known_cost": 784, "cost": 860, "family": "A", "name": "A-n32-k5", "optimality_gap": 0.096939}]

Judge Appendix

Summary of Replay-Aware CVRP Benchmark Suite Results

Condition	Replay Mode	Code Novelty	Final CVRPLIB Gap	Synthetic Holdout Gap	Transfer Gap	Adaptation Efficiency	Compression Pressure
cvrp_no_replay	none	0.0	0.2452	0.0124	0.1417	0.0	no
cvrp_random_replay	random	0.82	0.2426	**0.0054**	0.1371	0.0444	no
cvrp_failure_replay	failure	0.0	**0.2410**	**0.0054**	**0.1363**	0.0	no
cvrp_failure_replay_compression	failure + compression	0.0	0.2455	0.0054	0.1388	0.0	yes

Key Interpretations

- **Best Hold-Out Performance:**
- *cvrp_failure_replay* achieves the lowest held-out CVRPLIB gap (0.2410), indicating superior generalization on standard benchmarks.
- **Best Synthetic Holdout Gap:**
- Both *cvrp_failure_replay* and *cvrp_random_replay* achieve the lowest synthetic holdout gap (0.0054), suggesting strong performance on synthetic unseen distributions.
- **Best Overall Transfer:**
- *cvrp_failure_replay* leads in mean transfer gap (0.1363) closest to zero, demonstrating the best transfer ability between domains.
- **Code Novelty vs Transfer:**
- *cvrp_random_replay* shows high code novelty (mean 0.82) but offers only a marginal improvement in transfer metrics compared to *cvrp_no_replay*.
- *cvrp_failure_replay* achieves better transfer metrics without any increase in code novelty, indicating that transfer improvements arise from replay strategy rather than new algorithmic components.

- **Compression-Aware Replay:**
- Adding compression pressure (`*cvrp_failure_replay_compression*`) does not improve gaps and slightly worsens the held-out CVRPLIB gap and transfer gap relative to `*failure replay*` without compression.
- **No Replay Condition:**
- The baseline without replay (`*cvrp_no_replay*`) has the highest CVRPLIB gap and transfer gap, confirming that replay aids performance and transfer.
- **Adaptation Efficiency:**
- Only `*cvrp_random_replay*` exhibits nonzero adaptation efficiency (0.0444), but without surpassing `*failure replay*` in transfer metrics.

Conservative Conclusions

- Replay modes improve optimality gaps and transfer metrics over no replay.
- Failure replay leads to the best transfer performance and lowest held-out CVRPLIB gap with no reliance on increased code novelty, indicating effective knowledge reuse of failure cases.
- Random replay improves synthetic gap slightly and increases code novelty but does not outperform failure replay in transfer.
- Compression-aware failure replay does not deliver measurable improvement and may slightly degrade performance.
- Improvements are primarily algorithmic due to replay mode rather than lexical/code novelty.

Recommendation

Prioritize **failure replay** as the effective strategy for improving transfer and held-out performance in CVRP benchmarks while maintaining code simplicity and complexity. Random replay introduces more code novelty but has unclear benefits over failure replay. Compression-aware replay appears suboptimal in this setting.